

Gene regulatory guidance for interpretable single-cell trajectory inference in latent space

First Author^{1,2} and Second Author^{1,2}

¹First affiliation

²Second affiliation

Abstract

Time-series single-cell RNA sequencing observes population snapshots, but the cell lysis required for sequencing prevents individual cells from being followed over time. Schrödinger bridges infer continuous stochastic transport between these snapshots, often in latent representations that provide denoising but make the regulatory meaning of the learned dynamics difficult to interpret. We introduce Gene-Regulatory Alignment of Schrödinger Bridges (GRASB), a gene-regulatory-network-guided unbalanced Schrödinger bridge that couples gene-resolved regulatory information to latent transport. To solve this, we define a cell-contextualized transcription factor–target velocity in gene space, map it through the encoder of a frozen variational autoencoder using a Jacobian–vector product to guide the latent bridge, and map the learned drift back through the decoder. To test whether the resulting alignment depends on regulatory topology, we compare GRASB with matched unguided and degree-preserving shuffled-network controls. In the datasets tested, genuine network guidance yields decoded-drift genes that are more enriched for the corresponding cell-type and lineage programs than both controls. The results reveal a useful trade-off for applications that prioritize biological interpretability over distributional fidelity. Code is available at [\[repository URL\]](#).

1. Introduction

Single-cell RNA sequencing (scRNA-seq) measures cells destructively, so time-series experiments observe different cells at each timepoint. Trajectory-inference methods recover developmental order and topology from transcriptional similarity (Saelens et al., 2019), while optimal transport methods couple population snapshots and can account for growth and death (Schiebinger et al., 2019). Schrödinger bridges (SBs) extend this population view to distributions over continuous stochastic paths. Neural solvers have made this formulation practical for high-dimensional cell dynamics, including unbalanced settings in which population mass changes over time (De Bortoli et al., 2021; Pariset et al., 2023; Alatar and Wang, 2025; Zhang et al., 2025).

Because gene-expression measurements are high dimensional, these models often learn transport in a denoised latent representation. This reduces computational cost, but the representation combines signals from many genes. A decoder can map latent drift back to named genes, yet a recognizable gene ranking does not show that the dynamics follow regulatory programs specific to a cell type or lineage. Moreover, local decoded drift and generated expression answer different questions: drift is an instantaneous direction, whereas generated states integrate drift, diffusion, and population

This manuscript describes ongoing work.

birth or death. Biological organization of the vector field may therefore differ from fidelity of the reconstructed population.

Gene regulatory networks (GRNs) provide a complementary source of biological structure by preserving explicit transcription-factor (TF)–target relationships in gene space (Aibar et al., 2017). Context-specific linear models can assign signed coefficients to candidate edges and thereby define a regulatory direction at an observed cell state (Kamimoto et al., 2023). Combining these models creates a representational mismatch: applying a GRN directly to latent features discards gene identity, whereas learning transport in full expression space sacrifices the advantages of the latent representation.

We bridge the two representations with Gene-Regulatory Alignment of Schrödinger Bridges (GRASB), which transports a regulatory velocity rather than the network itself. A frozen VAE encoder maps a cell-contextualized gene-space velocity into the latent bridge through a Jacobian–vector product (JVP); after training, a decoder JVP maps latent drift back to signed gene velocities. The GRN acts as a soft prior rather than as causal ground truth. We test the method in human pancreatic endocrine differentiation and zebrafish embryogenesis (Veres et al., 2019; Farrell et al., 2018). Genuine GRN guidance improves alignment of decoded drift with lineage programs relative to unguided and degree-preserving shuffled-network controls, but worsens held-out population fit. These results motivate separate evaluation of local vector-field biology and integrated trajectory fidelity.

2. Related work

2.1 Schrödinger bridges for cell dynamics

An SB finds a path measure that matches specified endpoint distributions while minimizing relative entropy to a reference stochastic process (Schrödinger, 1932; Léonard, 2012). For Brownian reference dynamics, its static form is entropy-regularized quadratic OT and converges to quadratic OT as diffusivity vanishes (Carlier et al., 2017). Computationally, iterative proportional fitting (IPF) enforces the endpoint marginals and reduces to Sinkhorn scaling for discrete measures (Cuturi, 2013). Sample-based extensions replace a fixed spatial grid with regression, likelihood, or importance-sampling updates (Bernton et al., 2019; Pavon et al., 2021; Vargas et al., 2021).

Neural diffusion SBs continue this progression by learning forward and reverse drifts from simulated paths, which enables minibatched training in high dimensions (De Bortoli et al., 2021; Chen, Liu, et al., 2021). Standard SBs nevertheless conserve mass, whereas cell populations change through proliferation and death. Unbalanced OT introduces mass variation (Chizat et al., 2018; Liero et al., 2018), and unbalanced Schrödinger bridges (uSBs) incorporate it into the stochastic path law through killing and time-reversed birth (Chen, Georgiou, et al., 2022; Pariset et al., 2023). Within single-cell analysis, Waddington-OT estimates couplings between sampled times using growth information (Schiebinger et al., 2019); ARTEMIS combines a VAE with a uSB for continuous multi-timepoint trajectories (Alatkar and Wang, 2025); and mean-field extensions add cell–cell interactions (Zhang et al., 2025). These models reconstruct population dynamics but do not explicitly require their local drift to follow known regulatory programs.

2.2 Regulatory models of cell-state change

A complementary line of work represents regulatory structure directly. SCENIC combines coexpression with motif evidence to infer TF regulons and score their activity in individual cells (Aibar et al., 2017; Van de Sande et al., 2020). CellOracle goes further toward dynamics by fitting state-specific

regularized models on candidate TF–target edges and using the resulting signed matrix to propagate perturbations through gene-expression space (Kamimoto et al., 2023). These methods preserve gene identity and contextualize regulatory structure, whereas latent trajectory models provide denoising and scalable transport. Our method connects these complementary strengths by mapping a gene-space regulatory direction into a frozen latent model and testing whether any improvement in decoded drift depends on genuine TF–target topology.

3. Methods

3.1 Problem formulation

Let $x \in \mathbb{R}^G$ denote a cell’s preprocessed expression over G modeled genes, observed in population snapshots at $\tau_0 < \dots < \tau_M$. We rescale developmental time to $s \in [0, 1]$:

$$s(\tau) = \frac{\tau - \tau_0}{\tau_M - \tau_0}.$$

Because cells are not paired across snapshots, neither cell-level correspondences nor the intervening velocity field is observed. GRASB therefore learns a continuous latent uSB and guides its drift with a GRN-derived velocity defined over named genes. As summarized in Figure 1, a frozen VAE maps this velocity into latent space and maps the learned drift back to gene space.

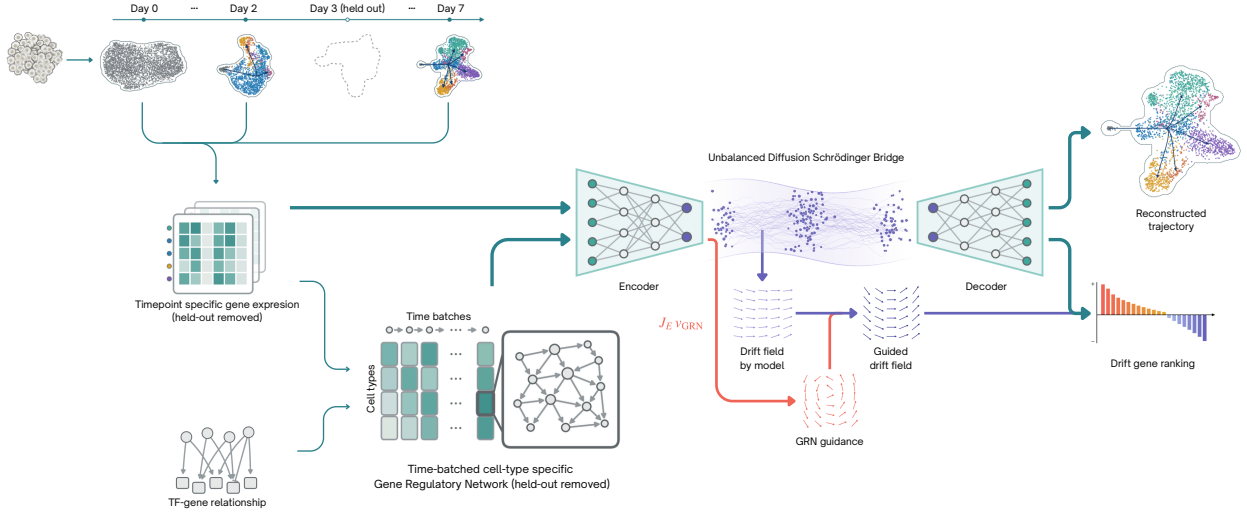


Figure 1: Overview of GRASB. A cell-contextualized regulatory velocity is mapped through the frozen VAE encoder to guide the latent drift, which is subsequently mapped through the frozen decoder for gene-resolved interpretation.

3.2 VAE-based denoising and dimensionality reduction

We first learn the latent representation. The VAE encoder $q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \text{diag } \sigma_\phi^2(x))$ and decoder $D_\psi : \mathbb{R}^d \rightarrow \mathbb{R}^G$ are trained by minimizing

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z|x)} \left[\|x - D_\psi(z)\|_2^2 \right] + \beta_{\text{VAE}} D_{\text{KL}}(q_\phi(z | x) \| \mathcal{N}(0, I)).$$

After training, the bridge uses the posterior mean $E(x) = \mu_\phi(x)$. The VAE is time-independent, trained only on the training timepoints, and then frozen so that every bridge condition, simulation, and readout uses the same gene-to-latent mapping.

To place the latent features on comparable scales, we standardize them using the training cells only. With training-set mean m_z and standard deviation r_z , the bridge state is

$$z' = S(E(x)) = \frac{E(x) - m_z}{r_z}, \quad S^{-1}(z') = m_z + r_z \odot z'.$$

Thus $\tilde{D}(z') = D_\psi(S^{-1}(z'))$.

3.3 Latent unbalanced diffusion Schrödinger bridge

3.3.1 Schrödinger-bridge transport

For endpoint distributions μ_0 and μ_1 , an SB seeks the path measure P^* closest in relative entropy to a reference diffusion R , subject to the prescribed marginals:

$$P^* = \arg \min_P D_{\text{KL}}(P \parallel R) \quad \text{such that} \quad P_0 = \mu_0, P_1 = \mu_1.$$

A diffusion SB learns neural forward and reverse drifts by alternating IPF regressions on simulated paths (De Bortoli et al., 2021). For a live particle, the forward process is

$$dZ_s = b_\theta(Z_s, s) ds + g(s) dW_s, \quad b_\theta(z, s) = f(z, s) + g(s)U_\theta(z, s),$$

where W_s is Brownian motion, f and g define the reference diffusion, and U_θ is a learned control. Forward and reverse controls use scalar-potential networks as in uDSB-F and ARTEMIS (Pariset et al., 2023; Alatkar and Wang, 2025).

For simulated increment ΔZ_k over Δs , the forward mean-matching loss is

$$\mathcal{L}_{\text{MM}}(\theta) = \mathbb{E}[\|\Delta Z_k - b_\theta(Z_{s_k}, s_k)\Delta s\|_2],$$

with an analogous loss for the reverse-time drift.

3.3.2 Unbalanced mass dynamics

The balanced construction above cannot represent changes in population size. To add this capability, uDSB-F augments $\mathbb{R}^d$ with a coffin state: forward killing becomes birth under time reversal (Pariset et al., 2023). Observed cell counts define relative masses, and a neural killing-rate correction is trained with the Ferryman loss to match cumulative birth and death at intermediate checkpoints, following ARTEMIS (Alatkar and Wang, 2025). The killing-rate model is optimized jointly with the bridge; drift and regulatory losses include only live particles.

3.4 Cell-contextualized regulatory velocity in gene space

We next construct the gene-space velocity used to guide the bridge. It combines a fixed candidate topology with context-specific signed coefficients.

3.4.1 Regulatory topology and coefficient estimation

For each species, a promoter-supported TF–target map defines a binary mask M , restricted to modeled genes and excluding self-edges (Kamimoto et al., 2023). It specifies candidate, not established causal, regulation.

Within this shared topology, context c combines a cell-type or lineage label with an interval between training timepoints. For target g , let $\mathcal{P}_g = \{j : M_{gj} = 1\}$. Target and regulator expression are centered within each endpoint, yielding $\tilde{y}_{g,c}$ and $\tilde{X}_{\mathcal{P}_g,c}$, and coefficients are estimated by

$$\hat{\beta}_{g,c} = \arg \min_{\beta} \left\| \tilde{y}_{g,c} - \tilde{X}_{\mathcal{P}_g,c} \beta \right\|_2^2 + \alpha_{g,c} \|\beta - \beta_{g,0}\|_2^2, \quad \alpha_{g,c} = \alpha_0 \left(1 + \frac{|\mathcal{P}_g|}{n_c} \right),$$

where n_c is the context size and $\beta_{g,0}$ is a training-only pooled estimate from the same lineage when sufficiently represented, or otherwise from the interval. The adaptive shrinkage stabilizes small contexts. Fits use only the two training endpoints; held-out times are excluded. Contexts with fewer than 30 cells receive no regulatory constraint.

The fitted coefficients define a sparse signed matrix $A_c \in \mathbb{R}^{G \times G}$:

$$(A_c)_{gj} = \begin{cases} (\hat{\beta}_{g,c})_j, & j \in \mathcal{P}_g, \\ 0, & j \notin \mathcal{P}_g. \end{cases}$$

Thus, the candidate topology is shared across contexts, while both the fitted coefficients and the expression state are context dependent.

3.4.2 Regulon velocity

For a gene-expression state x assigned to context c , the deployed regulatory velocity is

$$v_{\text{GRN}}^x(x, c) = \gamma_c A_c x.$$

Genes without fitted target rows receive zero velocity. Define the raw calibration velocity as $q_c(x) = A_c x$. For represented targets $\mathcal{C}_c$, the positive scale γ_c matches its mean magnitude to the observed rate of change between interval endpoints:

$$V_{\text{obs},c} = \frac{\left\| \bar{x}_{\tau_c^+, \mathcal{C}_c} - \bar{x}_{\tau_c^-, \mathcal{C}_c} \right\|_2}{\tau_c^+ - \tau_c^-}, \quad V_{\text{raw},c} = \frac{1}{|\mathcal{S}_c|} \sum_{i \in \mathcal{S}_c} \|q_c(x_i)_{\mathcal{C}_c}\|_2,$$

where $\mathcal{S}_c$ is the training-cell calibration set for context c . We set

$$\gamma_c = \frac{V_{\text{obs},c}}{V_{\text{raw},c}},$$

and fix this value before bridge training. The resulting velocity is an observational soft prior; its coefficient signs describe associations rather than established causal effects.

3.5 Regulatory guidance and gene-resolved interpretation

3.5.1 Mapping regulatory velocity to latent space

The regulatory velocity must now be compared with a drift learned in latent space. At latent state z' , let $\hat{x} = \tilde{D}(z')$. We transport the gene-space velocity as a tangent vector through the encoder:

$$v_{\text{GRN}}^z(z', s, c) = J_{S \circ E}(\hat{x}) v_{\text{GRN}}^x(\hat{x}, c) = \text{diag}(r_z^{-1}) J_E(\hat{x}) v_{\text{GRN}}^x(\hat{x}, c).$$

This and subsequent products are evaluated as JVPs without materializing Jacobians.

Before applying the penalty, we convert velocity from developmental time to normalized time by multiplying it by $\tau_M - \tau_0$, assign each bridge step to its developmental interval, and carry the source cell label along the path. Missing or unlabeled contexts receive no penalty.

The primary training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MM}} + \lambda_{\text{GRN}} \mathcal{L}_{\text{GRN}} + \mathcal{L}_{\text{mass}},$$

where $\mathcal{L}_{\text{mass}}$ denotes the Ferryman mass-matching terms and

$$\mathcal{L}_{\text{GRN}} = \mathbb{E}_{(Z_s, c) | \text{alive}} \left[\|b_\theta(Z_s, s) - v_{\text{GRN}}^z(Z_s, s, c)\|_2^2 \right].$$

Setting $\lambda_{\text{GRN}} = 0$ gives the matched latent uSB baseline.

3.5.2 Mapping learned dynamics to gene space

The inverse readout uses the same fixed representation. After training, the decoder differential maps the forward latent drift back to gene space. For a latent state z' ,

$$b_\theta^x(z', s) = J_{D_\psi}(S^{-1}(z')) \text{diag}(r_z) b_\theta(z', s) = J_{\tilde{D}}(z') b_\theta(z', s).$$

Its components are local signed gene-expression rates. We average them within each eligible lineage and time before ranking genes. Generated expression is evaluated separately by simulating from a source marginal and decoding the terminal states; it accumulates drift, diffusion, and birth/death rather than measuring the local vector field.

3.6 Matched controls

To distinguish topology-specific guidance from generic regularization, we compare the genuine-GRN condition with two matched controls. The no-GRN baseline sets $\lambda_{\text{GRN}} = 0$. The shuffled-GRN control preserves each target’s number of incoming edges, fitted coefficient values, context structure, and scale factor, but reassigns its regulators from the same TF pool while excluding self-regulation. The same reassignment is used across contexts. The shuffle therefore preserves the sparsity and coefficient distribution of the guidance term while disrupting TF–target identity. All three conditions use the same preprocessing, frozen VAE, uSB, mass-model architecture, and optimization settings; the bridge is trained independently for each condition.

4. Evaluation and results

4.1 Experimental design and evaluation criteria

We evaluated held-out intermediate timepoints in stage-5 human pancreatic differentiation (Veres et al., 2019) and zebrafish embryogenesis from 3.3 to 12 hours post fertilization (hpf) (Farrell et al., 2018). Pancreatic days 3 and 6 and zebrafish 5.3, 7, and 9 hpf were excluded from VAE, bridge, and GRN fitting. The primary zebrafish analysis used 7 and 9 hpf; 5.3 hpf was reserved for sensitivity analysis because its adjacent source contained fewer than 40 cells in two target lineages. Within each context, the genuine and shuffled arms used the same pre-calibrated γ_c . Model configurations and shared hyperparameters are summarized in Table 1.

Table 1: Model configurations and hyperparameters in the two developmental systems.

	pancreatic differentiation	zebrafish embryogenesis
VAE KL weight, β_{VAE}	0.3	0.3
guidance weight, λ_{GRN}	1	0.03
matched arms	no GRN, genuine GRN, degree-preserving topology shuffle	
latent representation	10-dimensional frozen VAE	
reference process	zero drift; triangular diffusion, $g_{\text{max}} = 0.5$	
numerical integration	100 Euler-Maruyama steps	
bridge architecture	hidden width 512	
optimization	batch size 512; learning rate 10^{-4}	
training schedule	six phases; ten epochs per phase	
trajectory updates	ten per direction and epoch	
path-batch reuse	five updates per simulated batch	

We evaluate the two model outputs separately. Decoded-drift metrics test whether the local vector field prioritizes an expected lineage program. Wasserstein-2 distance (W2), for which lower is better, instead compares generated and observed held-out populations after integrating the stochastic dynamics. The two evaluations therefore measure local biological organization and population reconstruction, respectively.

During model development, expression specificity proved insufficient on its own: a latent no-GRN model could produce lineage-specific expression signatures without recovering canonical markers. We therefore based the final evaluation on predefined gene sets and topology shuffles.

Pancreatic sets were the positive identity markers from Veres et al. (2019), and zebrafish sets mapped the six-somite modules from Farrell et al. (2018) to terminal lineages. Because these sets define expressed lineage identities rather than repressed programs, primary analyses used positive drift. GSEA normalized enrichment score (NES) evaluated each set against the complete signed drift ranking (Subramanian et al., 2005). Recall@ K measured the fraction recovered among the strongest positive-drift genes ($K = 15$ for pancreas; $K = 50$ for the broader zebrafish modules). Scoring every eligible lineage set against each ranking also tested whether the matching program ranked above alternatives.

4.2 Lineage-marker recovery through genuine GRN topology

We first evaluated pancreatic differentiation. Genuine GRN guidance improved both marker-set enrichment and individual-marker recovery over no GRN and all six shuffled topologies (Table 2).

Table 2: Pancreatic lineage-marker recovery and held-out population reconstruction.

	no GRN	genuine GRN	shuffled GRN
GSEA NES for the matching marker set	0.053	1.260	-0.158 ± 0.241
marker recall@15	9.4%	31.9%	$10.5 \pm 5.4\%$
W2, day 3	10.513	11.151	11.279 ± 0.176
W2, day 6	8.942	10.923	10.937 ± 0.051

* For GSEA NES and marker recall@15, higher values are better; for W2, lower values are better. Shuffled-GRN values are mean $\pm$ SD across six fixed randomized topologies.

The improvement was substantial: genuine guidance increased NES from 0.053 to 1.260 and recall@15 from 9.4% to 31.9%. Degree-preserving shuffles returned recall near the unguided level and produced negative mean NES, showing that the gain depended on regulatory edge identity. The matching marker set also ranked 2.89 on average, versus 5.27 after within-stage label permutation (one-sided $p = 0.0002$), indicating lineage-specific rather than global enrichment.

At the gene level, genuine guidance raised the positive-drift ranks of *NEUROG3*, *PAX4*, and *FEV* in endocrine progenitor states, as well as *SST*, *ARX*, and *DDC* in later endocrine states. These genes were expressed in their matched observed populations (Figure 2a,c), consistent with the lineage-specific marker-program enrichment.

4.3 Context-specific coefficient swapping

To test whether the fitted coefficients were interchangeable across states, we reassigned coefficient sets while keeping the common topology. From the lowest to highest quartile of expression divergence, marker recall for the evaluated state fell from 0.140 to 0.026, and the fraction of signatures matching neither the evaluated nor coefficient origin lineage rose from 63% to 87%. Thus, swapping coefficients between similar states preserved more of the original signal, whereas swaps between divergent states increasingly disrupted lineage-specific drift.

4.4 Lineage-module recovery in zebrafish embryogenesis

We next asked whether the pancreatic result generalized to zebrafish. Using the lineage modules of Farrell et al. (2018), genuine guidance again improved agreement over both controls (Table 3). These sets contained 21–51 genes, including 11–34 lineage-specific genes.

Across all three paired seeds, genuine GRN improved NES, recall@50, and matching-module rank over both controls. The advantage arose mainly at 9 hpf; at 7 hpf, the frozen VAE already supported strong module-aligned drift without guidance.

At the gene level, genuine guidance elevated somitic, adaxial, and axial mesoderm genes that the no-GRN model under-prioritized, including *RIPPLY1*, *CDKN1CA*, *MYOD1*, *PCDH8*, *TA*, and *TBX16* (Figure 2b). Observed expression placed these genes in the corresponding somitic, adaxial, or axial mesoderm populations (Figure 2d), matching the lineage assignments used for evaluation.

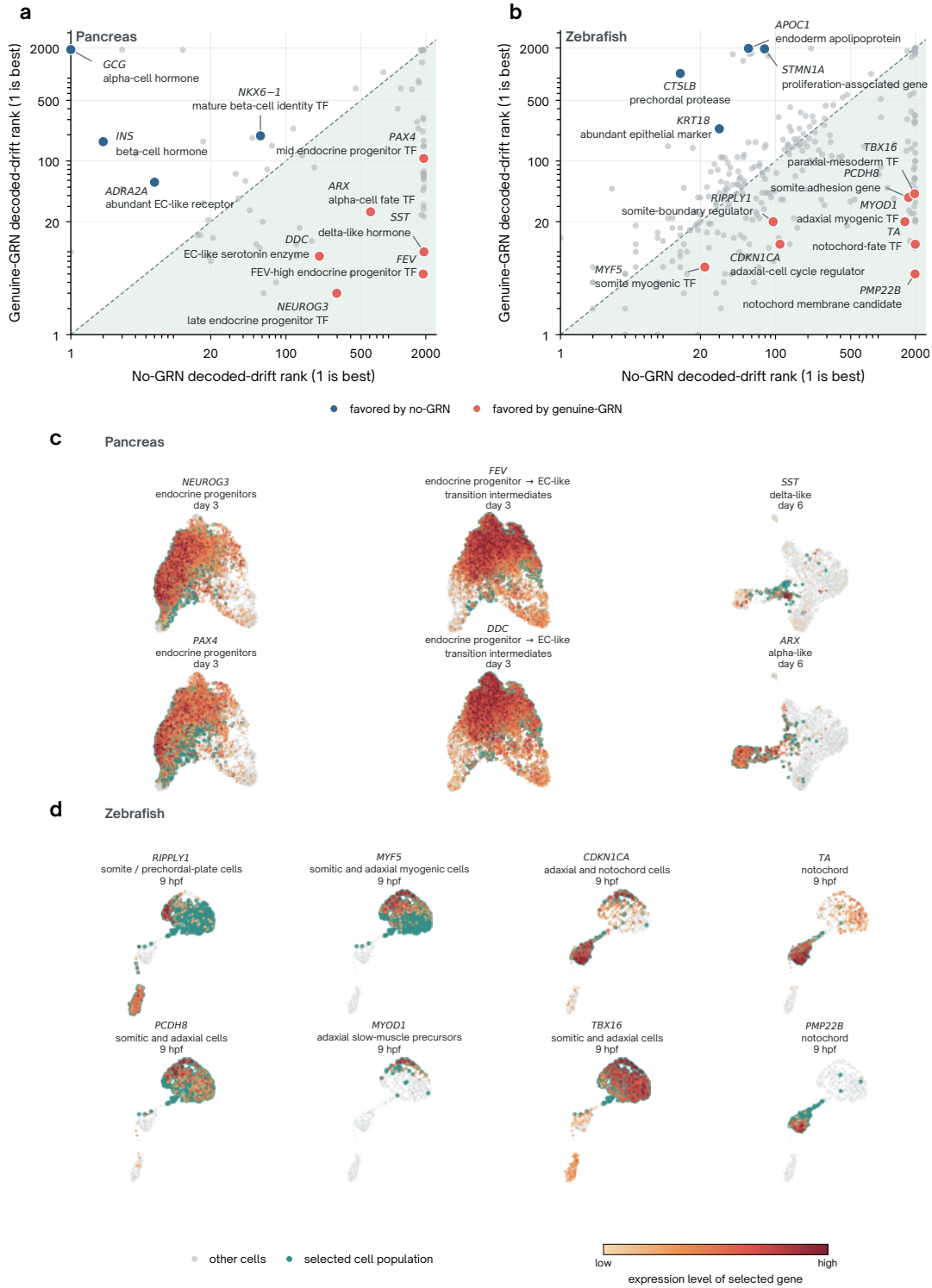


Figure 2: GRN guidance reprioritizes lineage-relevant decoded drift. Gene-rank comparisons show regulatory or lineage-program genes promoted from weaker no-GRN ranks and prominent output or state-associated genes shifted to weaker ranks by genuine guidance (a,b). Observed expression of guidance-promoted genes is shown in the matched cell populations (c,d).

Table 3: Zebrafish decoded-drift results at the primary 7 and 9 hpf targets.

	no GRN	genuine GRN	shuffled GRN
GSEA NES for the matching lineage module	1.106 $\pm$ 0.247	1.622 $\pm$ 0.044	0.711 $\pm$ 0.336
marker recall@50	0.365 $\pm$ 0.064	0.503 $\pm$ 0.017	0.299 $\pm$ 0.066
rank of the matching lineage module	2.370 $\pm$ 0.178	1.615 $\pm$ 0.340	3.356 $\pm$ 0.550
W2, 7 hpf	14.644 $\pm$ 1.243	16.891 $\pm$ 1.329	16.284 $\pm$ 1.703
W2, 9 hpf	15.376 $\pm$ 0.847	18.301 $\pm$ 1.808	17.963 $\pm$ 1.652

* Higher values are better for GSEA NES and marker recall@50; lower values are better for module rank and W2. Values are mean $\pm$ SD across three independently trained models.

4.5 Regulatory reprioritization of an expression-informed baseline

The rank comparison in Figure 2 shows that the no-GRN model was not devoid of biological information. It assigned strong ranks to prominent terminal products and state-associated genes, including *GCG*, *INS*, *ADRA2A*, and *NKX6-1* in pancreas and *APOC1*, *STMN1A*, *CTSLB*, and *KRT18* in zebrafish. These genes indicate that the frozen VAE supplied an expression-informed baseline capable of recognizing conspicuous features of the observed state.

The expression-aligned analysis in Figure 3 clarifies the limitation of that baseline. Without GRN guidance, decoded drift often favored terminal products or state markers such as *INS*, *GCG*, and *CALB2*, which were highly expressed in their matched populations. This was valid lineage information, but it was weighted toward genes already prominent in the observed state.

The relationship was not a simple mapping from mean expression to rank. Visibly expressed regulatory or lineage-associated genes, including *FEV*, *DDC*, and *NEUROD1*, remained comparatively weak without guidance, while the genuine-GRN model elevated *NEUROG3*, *CDKN1C*, *MAFB*, and *ARX*. Conversely, *SCG5*, *ACVR1C*, *SLC30A8*, *GAL*, and *SCG2* received meaningful ranks from both models. The no-GRN failure mode was therefore incomplete regulatory attribution, rather than absent lineage signal or deterministic ranking by expression abundance. Together, Figures 2 and 3 show that genuine GRN guidance reorganizes an expression-informed baseline toward regulatory and lineage-program genes that it otherwise under-prioritizes.

4.6 Trade-off between decoded-drift interpretation and reconstruction fidelity

This improvement in local drift did not translate into better population reconstruction. In both systems, improved marker or module recovery coincided with higher held-out W2 (Tables 2 and 3). Genuine and shuffled GRNs had similar W2 despite different biological recovery. In the pancreatic dataset, varying λ_{GRN} from 0.25 to 2 did not recover the distributional fidelity of the no-GRN model; most of the loss occurred at the smallest nonzero weight. Under these settings, imposing a nonzero local-vector-field constraint was the main source of the trade-off between drift interpretation and population reconstruction, rather than the GRN topology or the precise guidance weight.

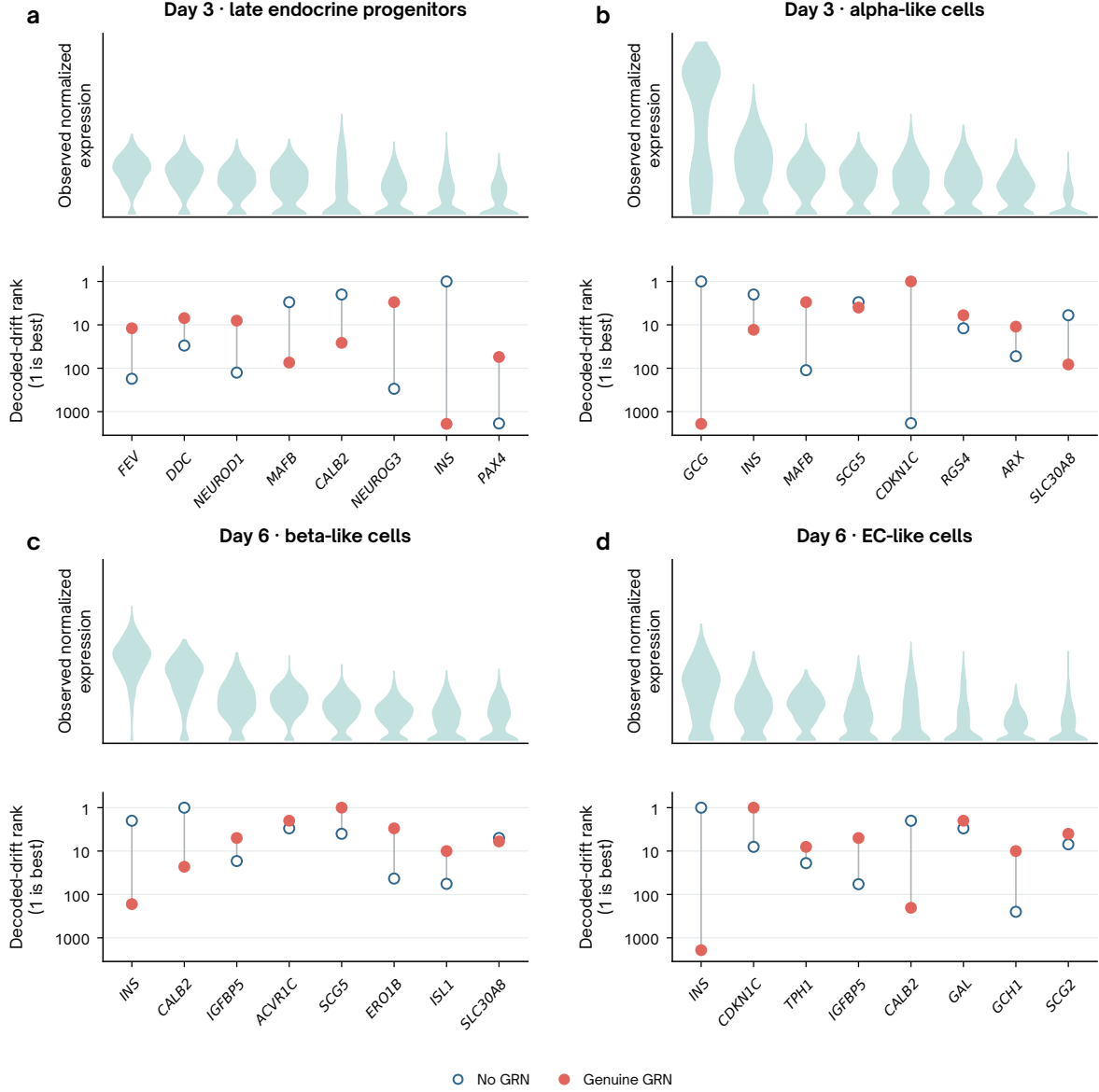


Figure 3: Expression-aligned ranks expose incomplete regulatory attribution without GRN guidance. Across four pancreatic lineage-day settings, no-GRN drift favors highly expressed output or state-marker genes while assigning weak ranks to several expressed regulators. Genuine guidance elevates those regulators; genes ranked strongly by both models show that the failure is incomplete regulatory attribution, not loss of lineage signal.

5. Discussion and limitations

GRASB couples gene-resolved regulatory information to latent stochastic transport without redefining the GRN over unnamed variables in the latent representation. The cell-contextualized GRN remains in gene space, where TF and target identities are preserved; an encoder JVP transports its velocity into the frozen VAE representation, and a decoder JVP returns the learned drift to signed velocities for named genes. Freezing the VAE makes this mapping stable across experimental conditions. The framework therefore combines the denoising and computational advantages of latent transport with a gene-level regulatory reference that can be directly tested.

The topology-controlled evaluation shows that this guidance changes the biological attribution of local drift. The frozen VAE baseline already recognized terminal identity genes and part of the shared lineage signal; genuine GRN guidance reorganized that signal toward regulatory and lineage-program genes that the baseline under-prioritized. This advantage was lost after degree-preserving edge shuffling, and fitted regulatory weights transferred best between transcriptionally similar states. Although this reprioritization is not a causal claim, it supports a specific interpretation: the GRN prior organizes the learned vector field toward known lineage programs through both regulatory edge identity and cellular context.

The results also separate vector-field interpretation from trajectory reconstruction. Guidance improved marker and module recovery but increased held-out W2, while genuine and shuffled GRNs produced similar W2 despite markedly different biological recovery. This pattern suggests that constraining the drift caused most of the fidelity loss, whereas genuine topology supplied the biological benefit. Consequently, distributional fit alone is insufficient when a trajectory model is used for gene-level interpretation, but marker alignment alone is also insufficient when accurate population generation is required. Applications should report both outcomes, select the guidance strength using training data or nested validation, and include a degree-preserving shuffled-GRN control to test whether interpretation depends on regulatory topology.

Several limitations bound these conclusions. We evaluated two developmental systems and one GRN-construction procedure based on observational expression and promoter-supported candidate edges. This prior does not capture distal enhancer activity, chromatin accessibility, or dynamic network rewiring; SCENIC+ and Dictys, along with CellOracle discussed above, provide important alternatives (Bravo González-Blas et al., 2023; Wang et al., 2023; Kamimoto et al., 2023). For each system, the cell labels and marker sets came from one source study, tying evaluation to that study’s definitions of cell identity. Comparisons across independent annotations, regulatory priors, and datasets are therefore needed to establish generality.

Measured perturbation responses provide the clearest next test of whether decoded drift predicts regulatory change. Time-resolved genetic or drug perturbations could evaluate response direction and heterogeneous trajectories, including the emergence of resistant states. Neural OT has already modeled single-cell drug responses from unpaired control and treated populations (Bunne et al., 2023), providing a relevant comparison. Extending the present framework to perturbation-conditioned transport would move the analysis from alignment with known developmental programs toward experimentally testable predictions of regulatory response.

References

- Aibar, S., C. B. González-Blas, T. Moerman, et al. (2017). “SCENIC: single-cell regulatory network inference and clustering”. In: *Nature Methods* 14.11, pp. 1083–1086. DOI: 10.1038/nmeth.4463.
- Alatkar, S. A. and D. Wang (2025). “ARTEMIS integrates autoencoders and Schrödinger Bridges to predict continuous dynamics of gene expression, cell population, and perturbation from time-series single-cell data”. In: *Bioinformatics* 41.Supplement_1, pp. i189–i197. DOI: 10.1093/bioinformatics/btaf218.
- Bernton, E., J. Heng, A. Doucet, and P. E. Jacob (2019). *Schrödinger Bridge Samplers*. arXiv:1912.13170 [stat.CO]. DOI: 10.48550/arXiv.1912.13170.
- Bravo González-Blas, C., S. De Winter, G. Hulselmans, et al. (2023). “SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks”. In: *Nature Methods* 20.9, pp. 1355–1367. DOI: 10.1038/s41592-023-01938-4.
- Bunne, C., S. G. Stark, G. Gut, et al. (2023). “Learning single-cell perturbation responses using neural optimal transport”. In: *Nature Methods* 20.11, pp. 1759–1768. DOI: 10.1038/s41592-023-01969-x.
- Carlier, G., V. Duval, G. Peyré, and B. Schmitzer (2017). *Convergence of Entropic Schemes for Optimal Transport and Gradient Flows*. arXiv:1512.02783 [math.AP]. DOI: 10.48550/arXiv.1512.02783.
- Chen, T., G.-H. Liu, and E. Theodorou (2021). “Likelihood Training of Schrödinger Bridge using Forward-Backward SDEs Theory”. In: *International Conference on Learning Representations*. URL: <https://openreview.net/forum?id=nioAdKCdXB>.
- Chen, Y., T. T. Georgiou, and M. Pavon (2022). “The Most Likely Evolution of Diffusing and Vanishing Particles: Schrödinger Bridges with Unbalanced Marginals”. In: *SIAM Journal on Control and Optimization* 60.4, pp. 2016–2039. DOI: 10.1137/21M1447672.
- Chizat, L., G. Peyré, B. Schmitzer, and F.-X. Vialard (2018). “An Interpolating Distance Between Optimal Transport and Fisher–Rao Metrics”. In: *Foundations of Computational Mathematics* 18.1, pp. 1–44. DOI: 10.1007/s10208-016-9331-y.
- Cuturi, M. (2013). “Sinkhorn distances: lightspeed computation of optimal transport”. In: *Proceedings of the 27th International Conference on Neural Information Processing Systems - Volume 2*. Vol. 2. NIPS’13. Red Hook, NY, USA: Curran Associates Inc., pp. 2292–2300.
- De Bortoli, V., J. Thornton, J. Heng, and A. Doucet (2021). “Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling”. In: *Advances in Neural Information Processing Systems*. Vol. 34, pp. 17695–17709. URL: <https://proceedings.neurips.cc/paper/2021/hash/940392f5f32a7ade1cc201767cf83e31-Abstract.html>.
- Farrell, J. A., Y. Wang, S. J. Riesenfeld, K. Shekhar, A. Regev, and A. F. Schier (2018). “Single-cell reconstruction of developmental trajectories during zebrafish embryogenesis”. In: *Science* 360.6392, eaar3131. DOI: 10.1126/science.aar3131.

- Kamimoto, K., B. Stringa, C. M. Hoffmann, K. Jindal, L. Solnica-Krezel, and S. A. Morris (2023). “Dissecting cell identity via network inference and in silico gene perturbation”. In: *Nature* 614.7949, pp. 742–751. DOI: 10.1038/s41586-022-05688-9.
- Léonard, C. (2012). “From the Schrödinger problem to the Monge–Kantorovich problem”. In: *Journal of Functional Analysis* 262.4, pp. 1879–1920. DOI: 10.1016/j.jfa.2011.11.026.
- Liero, M., A. Mielke, and G. Savaré (2018). “Optimal Entropy-Transport problems and a new Hellinger–Kantorovich distance between positive measures”. In: *Inventiones mathematicae* 211.3, pp. 969–1117. DOI: 10.1007/s00222-017-0759-8.
- Pariset, M., Y.-P. Hsieh, C. Bunne, A. Krause, and V. D. Bortoli (2023). “Unbalanced Diffusion Schrödinger Bridge”. In: *ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems*. URL: <https://openreview.net/forum?id=Vehxa7mBU9>.
- Pavon, M., G. Trigila, and E. G. Tabak (2021). “The Data-Driven Schrödinger Bridge”. In: *Communications on Pure and Applied Mathematics* 74.7, pp. 1545–1573. DOI: 10.1002/cpa.21975.
- Saelens, W., R. Cannoodt, H. Todorov, and Y. Saeys (2019). “A comparison of single-cell trajectory inference methods”. In: *Nature Biotechnology* 37.5, pp. 547–554. DOI: 10.1038/s41587-019-0071-9.
- Schiebinger, G., J. Shu, M. Tabaka, et al. (2019). “Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming”. In: *Cell* 176.4, 928–943.e22. DOI: 10.1016/j.cell.2019.01.006.
- Schrödinger, E. (1932). “Sur la théorie relativiste de l’électron et l’interprétation de la mécanique quantique”. In: *Annales de l’institut Henri Poincaré* 2.4, pp. 269–310. URL: https://www.numdam.org/item/AIHP_1932__2_4_269_0/.
- Subramanian, A., P. Tamayo, V. K. Mootha, et al. (2005). “Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles”. In: *Proceedings of the National Academy of Sciences* 102.43, pp. 15545–15550. DOI: 10.1073/pnas.0506580102.
- Van de Sande, B., C. Flerin, K. Davie, et al. (2020). “A scalable SCENIC workflow for single-cell gene regulatory network analysis”. In: *Nature Protocols* 15.7, pp. 2247–2276. DOI: 10.1038/s41596-020-0336-2.
- Vargas, F., P. Thodoroff, A. Lamacraft, and N. Lawrence (2021). “Solving Schrödinger Bridges via Maximum Likelihood”. In: *Entropy* 23.9, p. 1134. DOI: 10.3390/e23091134.
- Veres, A., A. L. Faust, H. L. Bushnell, et al. (2019). “Charting cellular identity during human in vitro β -cell differentiation”. In: *Nature* 569.7756, pp. 368–373. DOI: 10.1038/s41586-019-1168-5.
- Wang, L., N. Trasanidis, T. Wu, et al. (2023). “Dictys: dynamic gene regulatory network dissects developmental continuum with single-cell multiomics”. In: *Nature Methods* 20.9, pp. 1368–1378. DOI: 10.1038/s41592-023-01971-3.
- Zhang, Z., Z. Wang, Y. Sun, T. Li, and P. Zhou (2025). “Modeling Cell Dynamics and Interactions with Unbalanced Mean Field Schrödinger Bridge”. In: arXiv:2505.11197 [cs]. arXiv. DOI: 10.48550/arXiv.2505.11197.